

PRODUCT TECHNICAL DESCRIPTION

Material Code Number: **09:AP011**

SOUND-ABSORBING WALL PANELS

dukta® - flexible
wood

Technical Properties

- Usable Length (Standard): 2750 mm
- Usable Width (Standard): 863/892 mm (depending on the cut pattern)
- Thickness (Standard): 8 mm
 - Surface weight: Approx. 5 kg/m² (depending on construction and dimensions)
- Core Material: MDF natural | MDF color | Plywood | Three-layer panel (depending on the cut pattern)
- Minimum Bending Radius: Approx. 120 mm (depending on cut pattern, material type, and thickness)
- Optional: Acoustic fleece backing
- Other dimensions and shapes available on request.

Key Features

- * Excellent sound absorption
- * Consistently high sound absorption across all frequencies
- * Flexible
- * Fascinating aesthetics with different cut patterns
- * Versatile applications
- * Suitable for sound-absorbing wall and ceiling applications
- * Ideal for freestanding partition elements and design objects
- * Internationally patented process

Application Areas & Properties

- Our application recommendations in written and spoken form are based on our experience and the current state of science and practice.
- These recommendations are non-binding and do not establish any contractual legal obligation.
- Buyers are responsible for testing the suitability of our products for their intended purpose.
- Note: All test certificates, reports, and further technical information are available on request.

Acoustic Performance

Dukta elements achieve consistently high sound absorption values across all frequencies, unlike other absorber products that are effective only in specific frequency ranges.

Dukta acoustic elements are ideal for acoustically sensitive spaces. Their excellent acoustic properties, combined with captivating aesthetics, create high-quality spatial experiences.

Surface Finishes

Available Surface Options:

- Wood veneer (natural or lacquered)
- Clear or colored lacquer

Cut Patterns:

- Linar
- Sonar (vertical and horizontal)
- Janus (double-sided)
- Janus Tex (double-sided)
- Foli (lens-shaped)

Standard Cut: 4 mm groove / 4 mm web

Installation

- Dukta acoustic elements allow for quick and easy installation for permanent fixation.
- They can be attached to the substructure of acoustic walls using a pneumatic nailer or adhesive.
- Suitable for both flat and curved applications.

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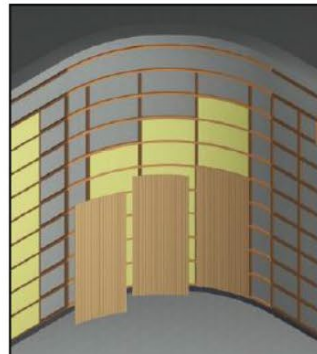
Pre-installation Conditions

Before starting the assembly, check that the conditions necessary for installation are met at the site (consult the conditions for TRANSPORT, STORAGE, INSTALLATION, and MAINTENANCE published on our website). To achieve the highest coefficient of sound absorption, create a space behind the panels, generally 200mm for false ceilings and 40mm for wall coverings, and place ISOVER ARENA APTA BASIC fiber with a density of 21 kg/m³ and a thickness of 48mm or equivalent supported on the back of the panel.



Assembly on Flat Surface

Create a well-leveled batten framework on the wall or ceiling to be covered, with a maximum distance of 400mm between battens. Place 45mm ISOVER ARENA BASIC fiber between the battens. Cover with an acoustic veil attached to the battens or the back of the panels with adhesive. Fix the panels on this batten base using adhesive and needles.



Curved Assembly

Create a wooden structure on the wall or ceiling, with the desired curves, observing the minimum curvature for each model. Maximum distance between supports is 400mm. Place 45mm ISOVER ARENA BASIC fiber between the battens. Cover with an acoustic veil attached to the battens or the back of the panels with adhesive. Fix the panels on this batten base using adhesive and needles.

Important: Most models can be installed on both walls and ceilings, except for thicker models like JANUS, which due to their weight and flexibility are not recommended for ceiling installation.

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Handling of Panels

- The panels are always delivered oversized, with a rigid frame around them to facilitate transport and handling. This frame has notches that facilitate the cutting position to achieve continuity of the panels and maximize their use. The technical sheets for each size and material indicate the dimensions of the raw panel and with the cut for maximum utilization. The cut must be made with a wood circular saw, adjusting the type of saw, depth, and feed to avoid splintering. Perform necessary tests before cutting the panels.
- The panels can be delivered natural (unfinished), lacquered, or varnished.
- If working with unfinished panels and wishing to apply the finish after installation, it is recommended to use thin waxes and stains for easier application on these types of panels.
- The handling, machining, and installation of the panels should be carried out with great care by experienced specialists in wood installations.



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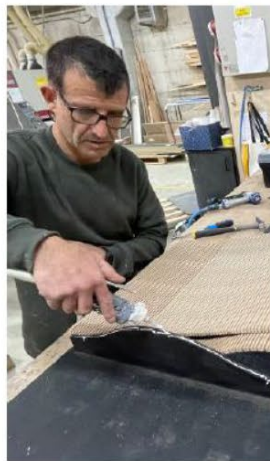
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Handling of Panels 2

- Once cut, the panels are very flexible in the direction perpendicular to the cuts. The technical sheets for each model and material indicate the minimum recommended bending radius. This radius is approximate and must be confirmed before carrying out the work.
- The curvature of the panels is provided by the supporting substructure that must be installed behind them. The panels are NOT self-supporting. They are a covering material.
- The panels are cut using a circular carpenter saw suitable for the type of work to be done.
- The panels are glued and nailed with 0.6mm needles onto the supporting structure.



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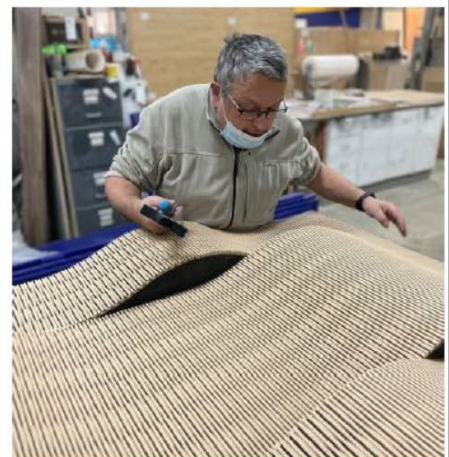
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Handling of Panels 3

- The panels are glued and nailed with 0.6mm needles onto the supporting structure.
- To achieve greater sound absorption, it is recommended to put 45mm ISOVER ARENA BASIC rock wool or equivalent in the space behind the panels.



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Handling of Panels 4

- To achieve greater sound absorption, it is recommended to put 45mm ISOVER ARENA BASIC rock wool or equivalent in the space behind the panels.
- A light fabric sheet (acoustic veil) should be installed between the panels and the structure to hide the mineral wool and structure that would be visible through the perforations of the panels.
- For most models, the acoustic veil can be requested separately or already adhered to the back of the panels. It is also possible to request the panels with the finish already applied (varnish, lacquer, ...).



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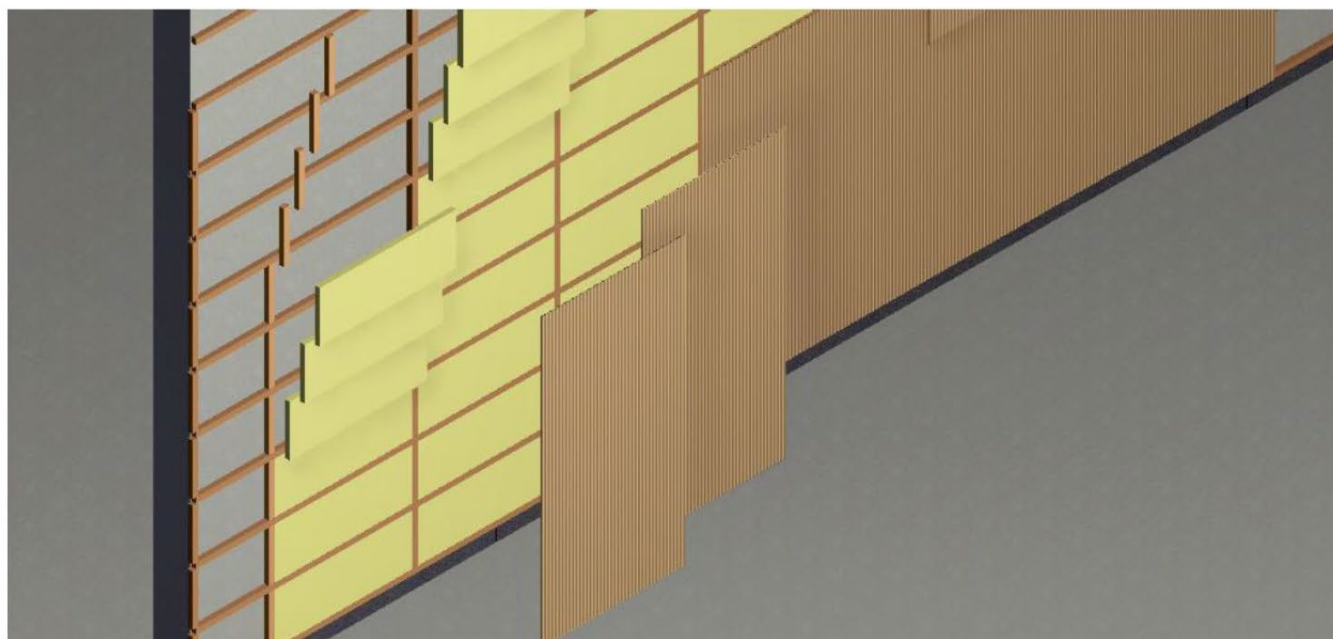
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Assembly on Flat Surface

- To mount on a flat surface, a substructure consisting of wooden battens should be installed. We recommend horizontally installed MDF battens of 40x45mm at a distance no greater than 400mm. Install segments of the same profile between the battens, aligning with the joints of the panels.
- For maximum sound absorption, it is recommended to install 45mm ISOVER ARENA BASIC rock wool or equivalent between the battens, behind the panels.
- The battens must be securely fastened to the wall and well-leveled. Avoid direct contact of the panels with the floor and protect the lower part using a baseboard.



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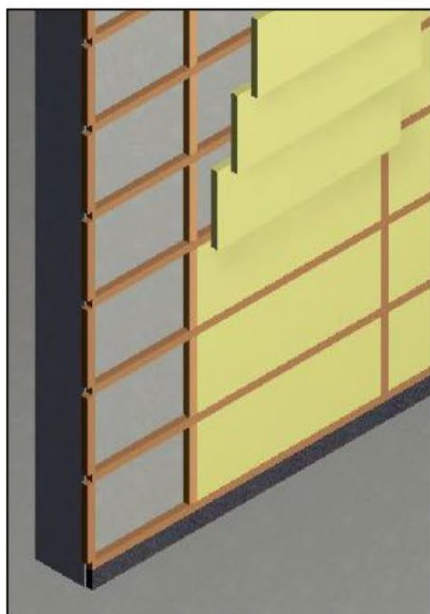
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Other general recommendations

The battens must be perfectly level and securely attached to the wall. The walls should not have any dampness.

Wooden panels should never be placed directly on the floor to avoid moisture problems. Protect the baseboards with skirting boards or other elements.

There are different types of hardware available to adapt to various mounting systems.



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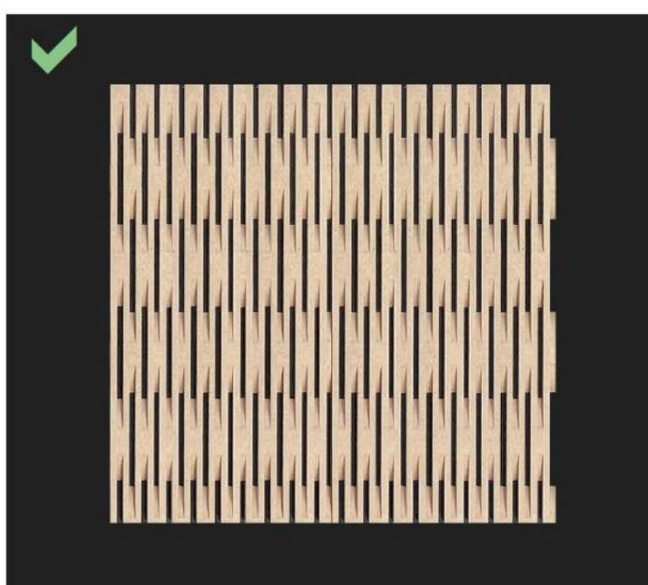
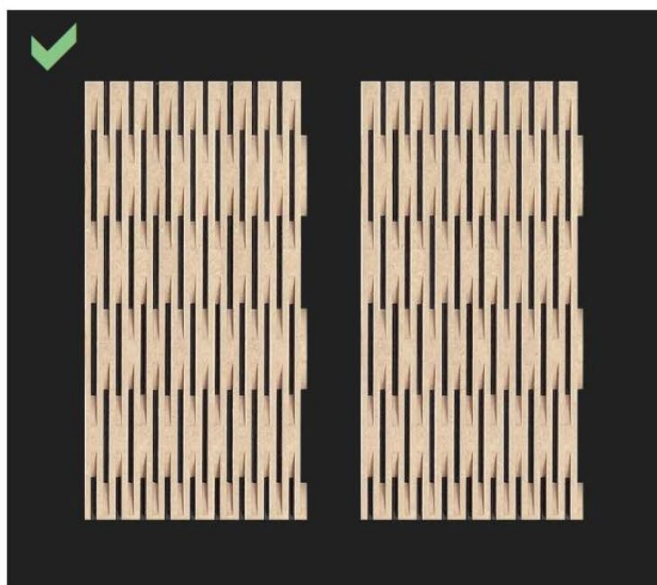
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Panel Cutting and Modulation

- Each DUKTA model has a repetitive grooving pattern throughout the panel. If the cuts and joints are made properly, the joints are practically invisible.
- Before making cuts, make sure they are made at the appropriate location with respect to the proper modulation.
- The images below reflect a properly made cut and joint.



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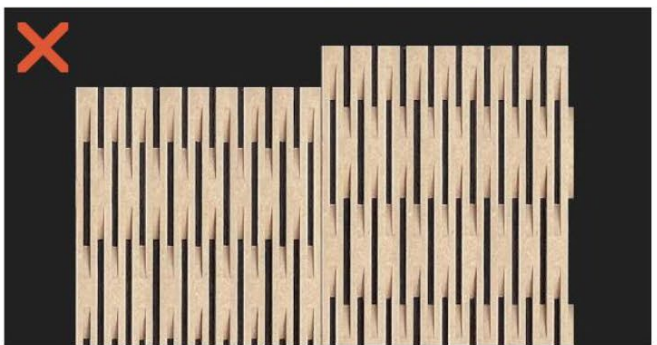
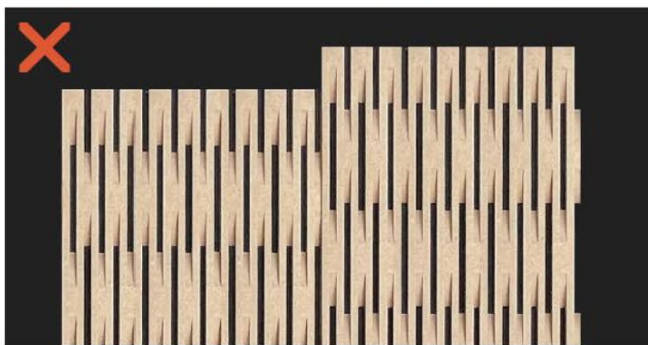
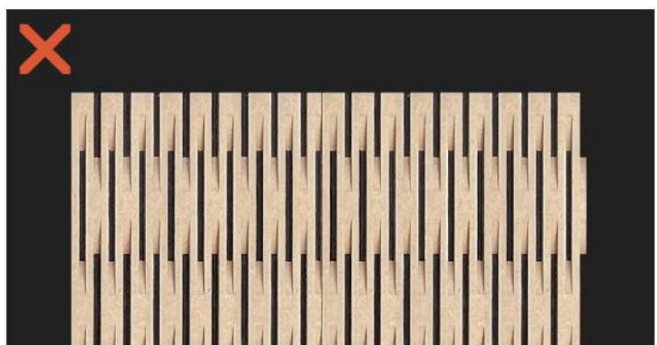
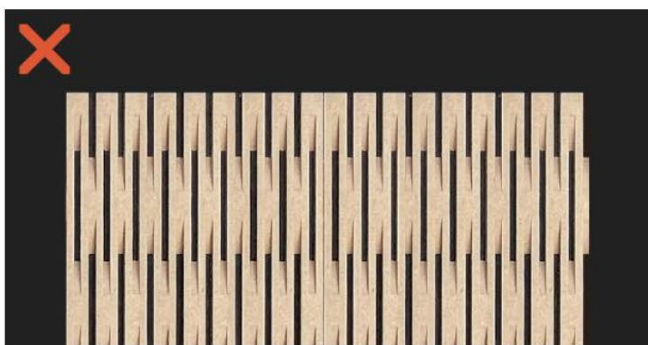
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Panel Cutting and Modulation

- The following examples show improperly made cuts and joints.



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Important

The images in this manual are only illustrative of the different installation concepts and recommendations; therefore the dimensions, scales and proportions shown may vary for a better understanding of the installation systems proposed. Illustrations have been completed with accessories not included.

Due to the high customization of our products, the information described in this documentation may vary.

The company is not responsible for eventual errors in writing, printing, technical data and translations.

This installation recommendations must be corroborated and adapted in each case by the technical management of each project.

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